

A wireless sensor network comprises a large number of wireless nodes that can communicate with each other and with a central host, transmitting data in real time. Some advanced sensors could even have a certain level of on-board processing capability to crunch bulky data into a single relevant figure and transmit only that to the host. The transmitted details could be about a player's position, a ball or bat's position, a player's heartbeat, energy level and so on. The information from various sensors would be collated and interpreted by the host computer for performance or game analysis.

Sensor networks could also be an umpire's aid, but more promising is their application in performance assessment and training, especially of a team as a whole.

A close parallel can be drawn from the European rugby teams' experiments with radio frequency identification (RFID) technology. The teams are experimenting with the use of RFID tags to transmit the exact coordinates of the ball and players at an astounding 2000 times per second. The information can be used to calculate movement, speed, accuracy, and even force of impact.

Incidentally, as far back as 2009, Apple patented a "network of sensors that deliver real-time velocity, impact, rotation and other data from sporting event participants to the Web." Let us see what application they came up with.

Wearable electronics

Nowadays, athletes are sporting a lot of electronics on their body, in their attire, accessories, etc, and not for frivolous reasons. Wearable electronics aids sportsmen mainly in fitness and performance analysis.

For some time now, Australian players are known to wear clothing that measures their heartbeat while training. Many such advanced suits for sportspersons are now easily available in foreign markets.

Last month, a company called Un-

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Since players' positions, speed, movement, etc are very critical to the success of most sports, GPS or location-based applications are becoming very popular in the sports arena.

The players would have to wear a thin band embedded with a chip, or simply have the chip stitched into their shirt, so that the coach can analyse their movements and positions in real-time using the relevant software applications. While in team sports this could help develop strategies and improve the game, in athletics it could help in performance analysis. If used wisely in games like cycling or motor-racing, it could even be life-saving as such location-information could help anticipate and prevent accidents.

Rapid prototyping to build skating boots

Speaking of wearable technology, here is an example of how a complementary technology is helping sportsmen design their own garments and accessories. Two experienced ice-skaters have recently developed a system that uses rapid prototyping to quickly design and develop boots. The system can be used to make custom-built shoes based on weather conditions, track conditions, their sporting style and so on.

derArmour launched a sports garment with an integrated electronics system. The garment contains a removable sensor pack near the breastbone and can give an athlete or coach instant feedback on breathing rate, heart rate, temperature and movements. The sensor pack contains a tri-axial accelerometer developed in conjunction with a Maryland company called Zephyr. The system measures acceleration and change of direction, which can be used to understand a player's performance. The data can be transmitted to wireless devices such as laptops and mobile phones. The company has now started launching these exclusively for selected teams, and by next year it will be available on store shelves.

In February, another company called SporTracker released an iPhone application that tracks a player's movement, speed, calories burnt, heartbeat and other critical fitness parameters using a thin arm-band. The information is gathered using built-in iPhone GPS and various sensors and transmitted in real time to a smartphone. The data

could also be relayed to friends using Twitter or Facebook.

Recently, some labs have also developed wearable devices for swimmers to self-assess their performance without the assistance of a coach. These devices usually consist of two bi-axial accelerometers positioned perpendicular to one another, and when worn on the lower back of the swimmer, can keep recording critical performance parameters for several hours. Lap times and stroke counts can be derived from the accelerometer data. Researchers are also trying to place sensors on the hands to monitor the propulsive forces.

The Sports and Biomedical Engineering Research group at Griffith University has come up with a motion-analysis tool for runners. The team has developed a tri-axial accelerometer based system using dual-axis ADXL202 accelerometers, a Hitachi microprocessor and other readily-available components. An on-board 10-bit analogue-to-digital converter manages sample rates of up to 250 Hz per channel and the data is encoded and transmitted using Bluetooth. Up to seven modules can be networked and synchronised. The client software acquires and analyses the data from various nodes worn on relevant limbs, to quantify aspects of elite running performance including ground contact time, impact attenuation, trunk stability and sprint start technique. In the near future, they propose to incorporate gyroscopes into the device to provide information about the orientation of the body under dynamic conditions.

A team of researchers in the Netherlands is working on a prototype